GEOMETRIC CONNECTEDNESS MAY BE CHECKED AFTER AN ARBITRARY FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations with mathematical macros, and of the project's own no-op definitions of the reference display environments; no macro was invented or redefined.

Editorial note. The cross-chapter standard prerequisite is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Geometrically connected scheme over a field. Same chapter, Definition environment defining a geometrically connected scheme as one whose base change to the algebraic closure stays connected. Supplies the notion being tested on both sides of the equivalence; the argument transporting it through the extension is the counted content.
- Existence of a common extension field of two given field extensions. Fields, Lemma labelled fields-lemma-common-extension-field (canonical Stacks tag 0H7K); invoked at proof lines 721-723. Supplies the field k"' containing both k' and an arbitrary extension k"; the surjectivity argument that transports connectedness back to k" is kept in the counted span.

Source reference. Original lemma 0H7K: sources/stacks-chow/tags.txt, source lines 21460–21460. Cross-chapter standard prerequisite invoked once at proof lines 721-723 for the existence of a common extension k"' of k' and k"; the surjectivity comparison of base changes that converts connectedness of X_{k"'} into connectedness of X_{k"} is retained verbatim in the counted span.

Lemma 0.1. *Let X be a scheme over the field k. Let k'/k be a field extension. Then X is geometrically connected over k if and only if $X_{k'}$ is geometrically connected over k'.*

Proof. If X is geometrically connected over k, then it is clear that $X_{k'}$ is geometrically connected over k'. For the converse, for any field extension k''/k there exists a common field extension k'''/k' and k'''/k'', see Fields, Lemma 0H7K. As the morphism $X_{k'''} \to X_{k''}$ is surjective (as a base change of a surjective morphism between spectra of fields) we see that the connectedness of $X_{k'''}$ implies the connectedness of $X_{k''}$. Thus if $X_{k'}$ is geometrically connected over k' then X is geometrically connected over k. □

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